# Mikhail Razakov

## Quantitative Research Candidate | C++ · Python

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Research-oriented engineer with professional compiler experience and hands-on C++/Python. I design controlled experiments, exact/reference oracles, randomized/metamorphic checks, and reproducible counterexample studies; my strongest evidence is in algorithms, program analysis, and systems research.

### Controlled experiments

UniversalToolchain: frozen corpus, valid controls, post-freeze holdouts, paired comparison, and repeated timing runs.

### Mathematical & algorithmic reasoning

PS-form: normalization, modular/GCD reasoning, exact affine analysis, and an independent exact oracle.

### C++ / Python research tooling

LLVM, Python harnesses, randomized/ metamorphic testing, reproducible counterexamples, and differential execution.

## EXPERIENCE

**October 2026 — present**

### ISP RAS — SharpChecker development

- Work on symbolic computation in SharpChecker, a C#/.NET static-analysis tool.

**July 1 — August 31, 2026**

### MCST — Compiler Engineering Intern

- Implemented an LLVM LICM pass in C++; transformation legality is derived from loop structure, memory accesses, side effects, and speculative-safety constraints.
- Built a Python harness comparing execution before/after optimization and checking explicit transformation / no-transformation cases; also implemented RPO, Dijkstra, Dinic, and Tarjan SCC in C++23.

## PROJECTS

### UniversalToolchain / Wist2

Implemented an end-to-end pipeline with frontend, IR, optimization passes, interpreter, and a CIL backend; current exact test manifest: 1,324/1,324.

### PS-form Memory Dependence Analyzer

Exact affine iteration-pair counting and modular/GCD disjointness proofs; yes/no cross-checked with a small-domain oracle and metamorphic tests, unresolved cases remain maybe.

### LLVM Interprocedural Global IV Optimization

A conservative LLVM 22 prototype/MVP with an actual IR transform; 29 positive/negative regression cases cover verifier validity, idempotence, and before/after execution.

## SKILLS

**Programming**

C++23, Python, C17, C#

**Experimental methods**

controlled experiments, exact/reference oracles, randomized/metamorphic testing, reproducibility

**Algorithms & CS**

data structures, graph algorithms, program analysis, CFG/SSA, compiler optimizations

**Research tooling**

Python harnesses, Linux, Git, CMake, counterexample analysis

## EDUCATION

HSE University — Software Engineering

## RECOGNITION

Organizer correspondence: only 5.0/5.0 score and only solution passing all 104/104 tests among 49 submissions.

Moscow / remote